

NVDRAM: A 32Gb Dual Layer 3D Stacked Non-volatile Ferroelectric Memory with Near-DRAM Performance for Demanding AI Workloads

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Abstract — We present NVDRAM, the world's first dual-layer, high-performance, high-density (32Gb) and non-volatile ferroelectric memory technology. NVDRAM uses an ultra-scaled (5.7nm) ferroelectric capacitor as memory cell and a dual gated, stackable, polycrystalline silicon transistor as access device. To achieve high memory density, two memory layers utilizing a 4F² architecture with 48nm pitch are fabricated above CMOS circuitry. Full package yield is demonstrated from -40°C to 95°C along with 10yr reliability (endurance and retention). A new component qualification flow based on JEDEC JESD47 that combines both DRAM and NVM tests is successfully implemented. NVDRAM utilizes the LPDDR5 command protocol and system compatibility is successfully demonstrated using a commercial development platform. NVDRAM achieves a bit density of 0.45Gb/mm², higher than Micron's industry-leading 1β planar DRAM technology.

I. INTRODUCTION

Since the slowing of Dennard's Scaling in the early 2000's, computational systems have increasingly relied on a number of heterogeneous system solutions to keep up with performance demands that general purpose CPUs have struggled to provide. This stratification of computing platforms has ultimately enabled new use cases such as Artificial Intelligence (AI) and Machine Learning (ML) to flourish. Critically, the rapid growth of model size in AI and ML has been an integral driver in the increasing usefulness and performance of these applications, with the latest explosion in model size coming from GPT-4 with more than one trillion (10¹²) model parameters. Over the last decade, model size has exhibited exponential growth and shows no signs of slowing (Fig. 1) [1-2]. Top-down growth of deep learning workloads has outpaced the underlying hardware, particularly the memory and storage sub-systems. Memory capacity expansion beyond what is viable for direct attached memory is required to maintain performance and power efficiency. Revolutionary architectures like near-memory compute or processing-in-memory remain nascent and will require a huge ecosystem lift, including software capability, to ultimately provide value and efficiency.

To enable increasingly complex, large model deep learning, near-DRAM performance and low energy are required. Outfitting existing, traditional compute architectures utilizing domain-specific acceleration with a more efficient memory hierarchy offers the best solution to improve data movement and enable increasingly large model size in the near term. Unfortunately, the current memory hierarchy offers only three tiers of cost-performance-energy capability represented by DRAM, NAND and HDD. Compute Express Link (CXL) provides a platform that can bridge the current memory hierarchy gap and allow for additional memory capacity and bandwidth expansion without the need to strictly adhere to

traditional DRAM timings (Fig. 2). The CXL memory interface abstraction and latency characteristics provide a near ideal opportunity to leverage NVDRAM's strengths in capacity and bandwidth while mitigating any negative impact from its longer latency compared to DRAM. NVDRAM achieves near-DRAM performance and low energy operation while simultaneously reducing cost structure compared to DRAM, enabling capacity growth while remaining economically viable. Non-volatility, although not explicitly required, is expected to be of long-term strategic value by simplifying compute architecture as well as offering opportunities for further optimization.

II. EMERGING MEMORY LANDSCAPE

Academia and industry have explored several emerging memories over many decades. Through extensive work, it has been proven that none of the studied technologies can directly replace DRAM from a product standpoint due to the strict latency, power and bandwidth requirements of DRAM. Longer latencies of Resistive RAM, Phase Change Memory and 3D XPoint limit their use to storage class memory. Both Spin Torque Transfer (STT) and Ferroelectricity (FE) based memories can achieve near-DRAM latencies (Fig. 3), but the scalability and capacity of STT is severely limited due to the high current requirements for cell operation and low sense margin of scaled STT cell architectures. FE-based NVDRAM technology is the only technology that provides the combined capacity, scalability, performance, power, reliability and cost to suit the requirements of high-capacity AI/ML applications requiring near-DRAM performance. Additionally, the non-volatile nature of NVDRAM allows for reduced refresh overhead commands relative to conventional DRAM, leading to power reduction and reduced memory access collisions. This will enable systems to increase memory capacity while avoiding classic refresh related issues thus allowing system designers to consider capacities of 128Gb or more.

III. DUAL LAYER NON-VOLATILE MEMORY

NVDRAM is the highest capacity FE memory to date (Fig. 4) [3-7]. To enable the high-density, high-performance requirements for a near-DRAM memory solution, a dual-layer, 1T1C (1-Transistor, 1-Capacitor) memory is developed for the first time. Stacking two memory layers achieves high bit density and lower cost structure compared to conventional DRAM but requires advanced integration solutions and the development of low temperature processes for both access device and memory cell. To maximize array efficiency (AE), the memory layers are fabricated on top of a high-performance CMOS capable of LPDDR5 speed targets. Each memory layer constitutes 16Gb and is configured like DRAM with orthogonal word-lines and digit-lines at 48nm pitch. Metal-insulator-metal decoupling capacitors are constructed on top of the memory array to further

improve AE. The final die picture and cross-section are shown in Fig. 5. At 48nm pitch and with CMOS under the array, NVDAM technology achieves a bit-density of 0.45Gb/mm², surpassing Micron's industry-leading 1 β planar DRAM technology.

Access Device - NVDAM utilizes polycrystalline silicon due to its high-performance and ease of manufacturability, while limitations in reliability of oxide semiconductors and the immature status of 2D semiconductors prevents their use. A 4F² dual gated transistor with vertical channel is fabricated from doped polycrystalline Si (Fig. 6). Nano-second pulsed Laser Annealing (LA) crystallizes the channel and activates the junctions. Optimization of LA process is key to increase grain size for high mobility, reduce dopant diffusion to achieve tail-free I_{OFF} distributions, and limit heat transfer to the underlying memory layer. To enable large arrays, high AE and low cost; Ruthenium is used as WL material due to its 1/5th resistance as compared to TiN. I_{ON} >30 μ A/device, I_{OFF} <200fA/device, and V_{TCC} = 0.2V are achieved at V_{GS}=3.4V and V_{DS}=1.5V (Fig. 7). NVDAM achieves highest I_{ON} amongst stackable devices reported in the literature at similar bias conditions and normalized for width using published data (Table. 1).

Memory Cell - The memory cell is a metal-insulator-metal capacitor comprised of doped HfZrO_x-based ferroelectric (5.7nm) with TiN-based electrodes (Fig. 8). Since read signal is directly proportional to the 2Pr of the ferroelectric cell, its value must be maximized throughout the life of the product (i.e., uncycled and cycled). Record-high *effective* 2Pr of >55 μ C/cm² at 1.5V operating voltage is achieved while cycling up to 10¹², with a peak value of 64 μ C/cm² (Fig. 9, Fig. 10). Effective 2Pr is defined as the minimum 2Pr maintained *throughout* the entire cycling range tested. Composition optimization, electrode engineering, interface treatments and crystallization anneals are all necessary to achieve record endurance and retention performance while maintaining stability through dual layer integration. Critically, the addition of dopants is needed to stabilize the orthorhombic phase and minimize tetragonal and monoclinic phases, resulting in high initial 2Pr, minimal wake-up and record high effective 2Pr amongst published results (Fig. 11). Switching kinetics are accurately modeled over a wide timescale (1ns-10s) using an internally developed Micron model. The ability to accurately model kinetics of both high-speed switching as well as slow thermal relaxation and various disturb phenomena is critical to enable high confidence product design simulations and architecture evaluations (Fig. 12b).

IV. COMPONENT PERFORMANCE

NVDAM is LPDDR5 compatible for quick adoption and system compatibility. Fig. 13a compares NVDAM to typical LPDDR5 array timings. Longer array sensing and array writeback contribute to the key parameter changes in tRCD and tRP. Fig. 13b shows the core operation waveforms for a NVDAM read/write cycle. The array operation is similar to DRAM, but a fixed reference is used during sensing. The cell plate is toggled to write-back one of the memory states during pre-charge, enabling a shorter tWR but increasing tRP. Because of similar array operation and the ability to use symmetric read and write conditions, NVDAM maintains the DRAM

command protocol with only slight timing modifications. REF commands are repurposed as Memory Maintenance (MM) commands; however, unlike DRAM, they are not required if there is no activity. MM commands are fully compatible with LPDDR5 RFM and used to achieve reliability and Bit-Error-Rate (BER) targets without impacting component speed. The MM overhead of NVDAM is significantly less than DRAM: ~20x less at 100% activity (worst-case) and ~150x lower at 10% activity (Fig. 14). Full system compatibility is demonstrated using a commercial development platform. NVDAM on the development platform is able to run internal benchmarks used for Micron's mobile DRAM for >100h without issue and successfully completes the glmark2 benchmark.

Yield and Reliability - Array-level performance of both memory layers is matched. Fig. 15a shows the distributions of ones and zeros data states for a representative yielding die with zero fails, achieving a BER <3 $\times 10^{-11}$ across both decks and exhibiting a memory read window of 750mV at -6.54 σ (32Gb). NVDAM shows excellent immunity to digit and plate disturbs as shown in Fig. 16a. This result is confirmed by the lack of fails after 100k row-hammers applied to a 1Gb array space (BER <10⁻⁹, Fig. 16b). Fig. 17 shows the performance of NVDAM across temperature (-40°C to 95°C). Wafer Level Reliability (WLR) tests are designed to monitor the performance over time while sampling a statistically significant distribution of the array. WLR testing flow is shown in Fig. 19. The retention results of a high temperature storage bake, equivalent to more than 10yr at 55°C, are shown in Fig. 18. The activation energy for imprint and retention are extracted (1.2 $\pm$ 0.1eV, Fig. 18 inset). Array distributions of ones and zeros after 10¹⁰ endurance cycles show excellent performance (Fig. 20).

Given the low latency, high endurance, and non-volatility of NVDAM; a new qualification flow based on JEDEC JESD47 combining DRAM and NVM tests and relevant cycling patterns is used to qualify NVDAM [8]. All reliability data is collected at high temperature (T_{Cyc}=95°C, V_{Cyc}=1.5V) where memory window is lowest. Fig. 21 shows the measured distributions from the Non-Volatile Cycling Endurance flow, demonstrating robust post-package reliability. NVDAM maintains a positive read window at BER of 3 $\times 10^{-11}$ after extrapolating to 10yr usage based on Micron's internal lifetime reliability model (Fig. 21 inset). Additionally, no die fails are observed for UCHTDR, PCHTDR and HTOL.

V. CONCLUSION

Combining the high endurance and non-volatile nature of the FE memory cell, NVDAM achieves DRAM-like read/write latency and endurance while surpassing the retention performance of NAND (Fig. 22). Using a dual layer architecture, NVDAM achieves record density and is an excellent choice for CXL and AI/ML applications requiring high-capacity and near-DRAM performance.

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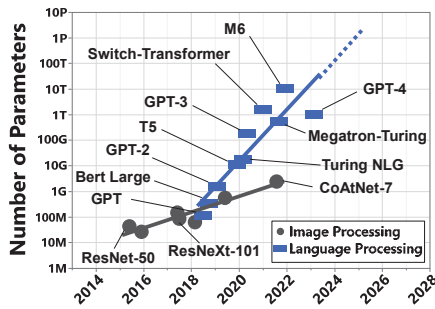


Figure 1. Number of Parameters in AI/ML models over time showing rapid growth in deep learning model size since 2018. Number of parameters quickly approaching one quadrillion (10^{15}).

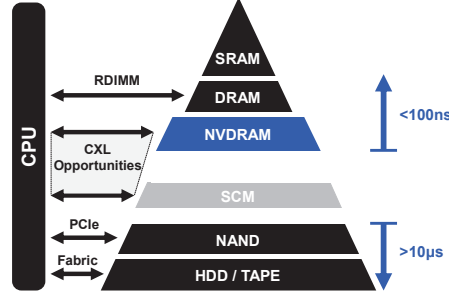


Figure 2. NVDRAM achieves near-DRAM performance with read latency $<100\text{ns}$. CXL provides a platform to enable new memories without strict adherence to DRAM timings.

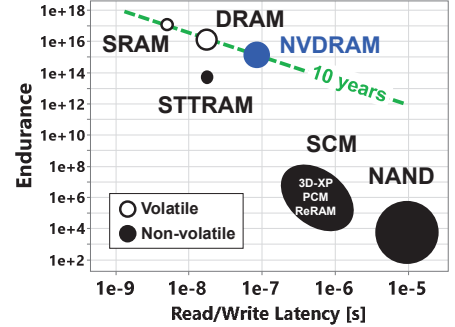


Figure 3. Cycling endurance versus latency. NVDRAM achieves near-DRAM latency and has sufficient cycling endurance to support 10yr product lifetime.

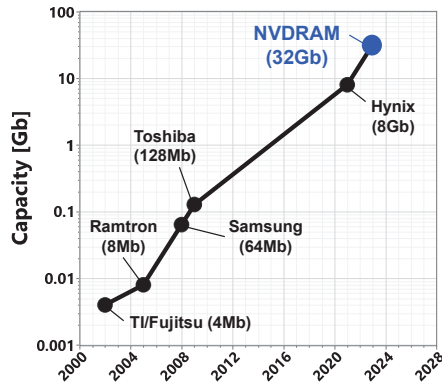


Figure 4. Capacity versus time for ferroelectric memories reported by various companies. NVDRAM uses 1T1C dual-layer memory to achieve 32Gb at $0.45\text{Gb}/\text{mm}^2$ bit density. [3-7]

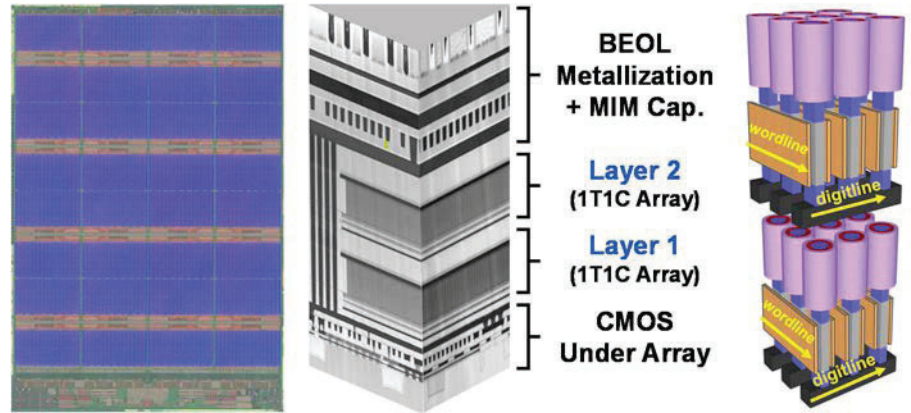


Figure 5. Final die layout (left) and SEM cross-section (center) of 32Gb NVDRAM showing 1T1C memory layers fabricated over CMOS array. Schematic diagram of NVDRAM memory arrays (right) showing polysilicon access device with orthogonal WL and DL and ferroelectric memory cells.

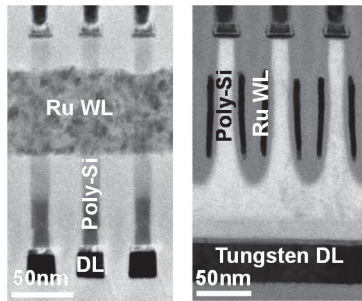


Figure 6. Access device cross-sections along WL (left) and along DL (right), showing poly-silicon access device with double-sided gate.

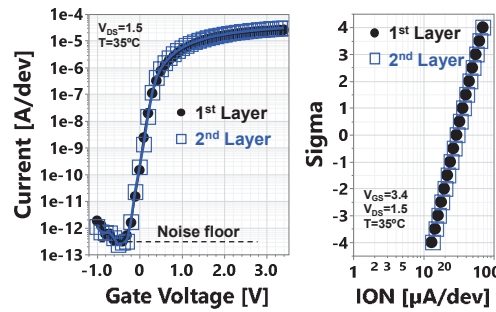


Fig. 7. Access device Current-Voltage (IV) characteristics for both layers at 35°C (left), showing matched device characteristics. I_{ON} distribution of 1st and 2nd layers to -4σ (right), showing well controlled log-normal distribution.

Parameter	Micron (This Work)	[1] TSMC (2020 IEDM)	[2] IMEC (2022 VLSI)	[3] IMEC (IEDM 2019)
Channel	Poly-Si	Poly-Si	IGZO	MoS ₂
VLSI + Low Thermal Budget Capable	Yes	Yes	Yes	No
Gate Stack [Å]	42.5 SiO ₂	30 SiO ₂ + 50 HfO ₂	50 Al ₂ O ₃	40 HfO ₂
I_{ON} ($\mu\text{A}/\mu\text{m}$)	133	53	24	27
I_{OFF} (fA/ μm)	442	889	0.0004	500
V_{t} [V]	0.20	0.31	0.3	0.3
SS (mV/dec)	79	82	100	80
Lg [nm]	80	50	40	100
W_{EFF} [nm]	40	90	80	135
$V_{\text{DS}}/V_{\text{DS}}$ [V]	1 / 1.5	1 / 1.5	1.3 / 1	1 / 1

[1] F.K. Hsueh, 2020 IEDM; [2] S. Subhechha, 2022 VLSI; [3] Q. Smets, 2019 IEDM

Table 1. NVDRAM access device compared to other low thermal budget access devices at similar bias conditions. I_{ON} and I_{OFF} are normalized using W_{EFF} (total channel width). I_{OFF} is measured at $V_{\text{DS}}=1\text{V}$.

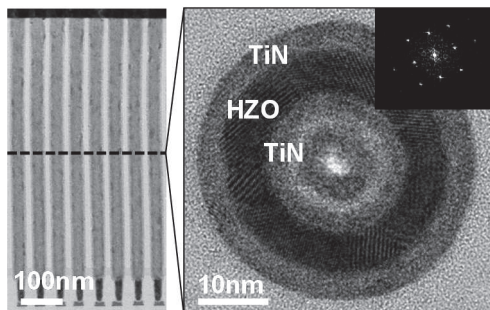


Figure 8. SEM cross-section of memory cells (left) and plan-view TEM showing TiN electrodes and crystalline ferroelectric (right).

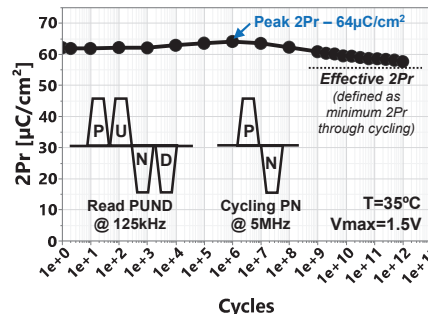


Figure 9. Polarization versus cycles up to 10^{12} (1.5V, 35°C). Read performed with $2\mu\text{s}$ pulses, cycling performed with 100ns pulses (5MHz).

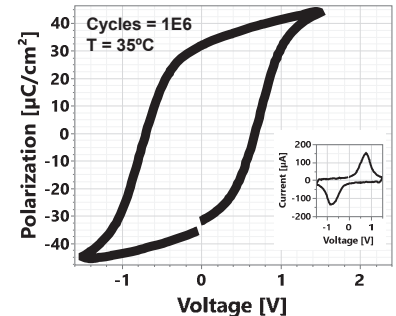


Figure 10. Polarization versus voltage (PV) and corresponding current versus voltage (IV) measured at 35°C after 10^6 cycles.

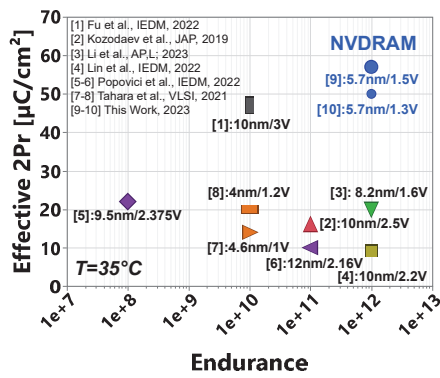


Figure 11. Effective 2Pr versus cycles. Effective 2Pr is defined as the minimum 2Pr throughout cycling as shown in Figure 9. NVDRAM achieves record high effective 2Pr due to its high starting 2Pr and low fatigue.

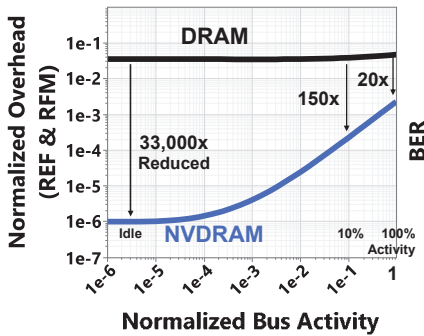


Figure 14. Combined refresh (REF) and refresh management (RFM) normalized overhead versus normalized bus activity.

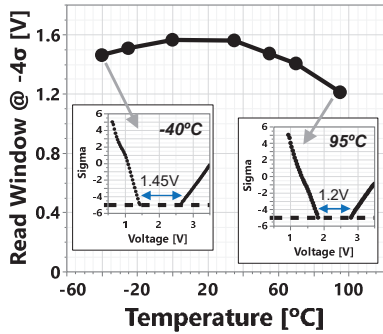


Figure 17. NVDRAM memory window versus temperature. Memory window is measured at -4σ from full array distributions (insets). No tails are observed at cold or hot.

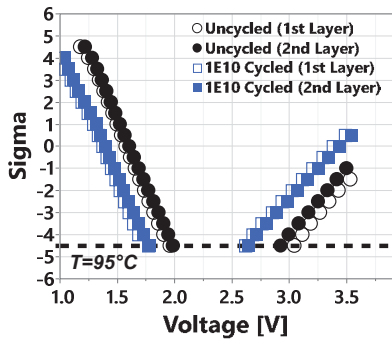


Figure 20. NVDRAM zero and one distributions for time zero (circles) and post 10^{10} cycles (squares). Both cycling and read were performed at 95 °C.

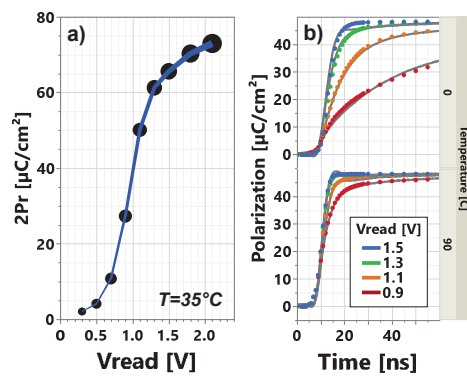


Figure 12. (a) 2Pr versus read voltage showing saturation in switching kinetics near 1.3V. (b) Overlay of Micron's FE kinetic model (dots) to high-speed polarization versus time data (lines) showing good fit across bias and temperature.

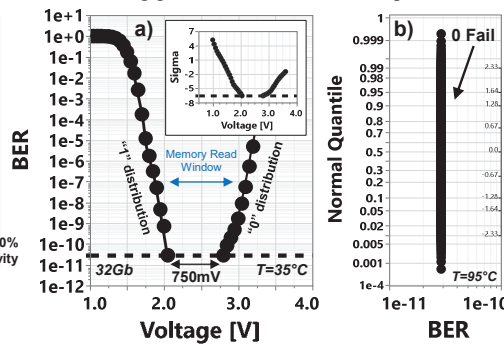


Figure 15. (a) Zero and one distributions for NVDRAM die at 35 °C. (b) BER distribution of 1k die, showing zero fails at 95 °C on all sampled die.

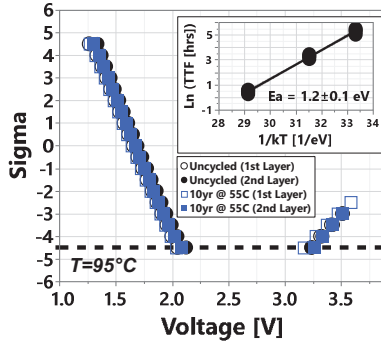


Figure 18. NVDRAM zero and one distributions for time zero and post 10yr bake at 55°C. Inset shows activation energy (1.2 ± 0.1 eV).

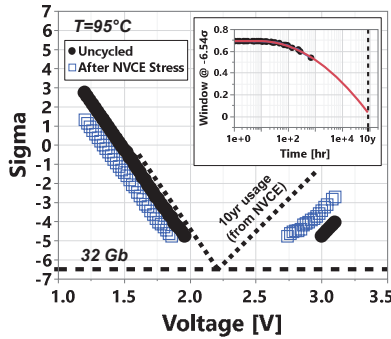


Figure 21. Zeros and ones distributions showing time zero, post NVCE stress and predicted after 10yr usage. Read window vs usage (inset) shows positive window after 10yr.

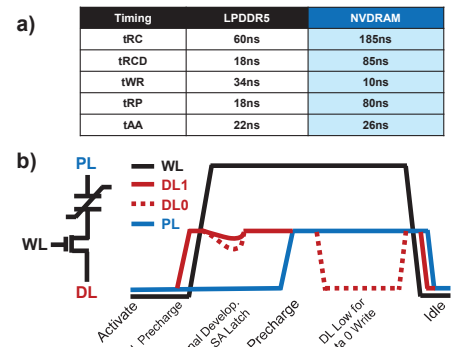


Figure 13. (a) Timing comparison to typical LPDDR5 DRAM. (b) Diagram of basic array operation. Data 1 shown with solid line. Data 0 shown with dashed line. Sensing is destructive to Data 0. Data 0 writeback occurs during precharge.

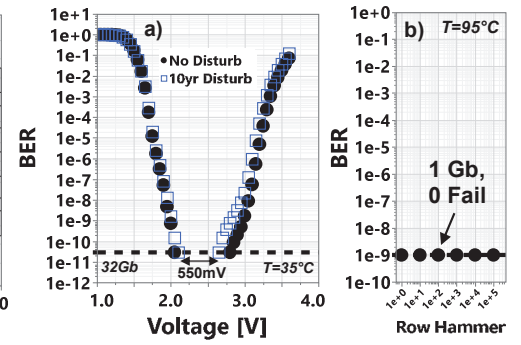


Figure 16. (a) Distributions showing before and after digit/plate disturbs. (b) No fails after 100,000 row-hammers applied to a 1Gb array space ($BER < 10^{-9}$).

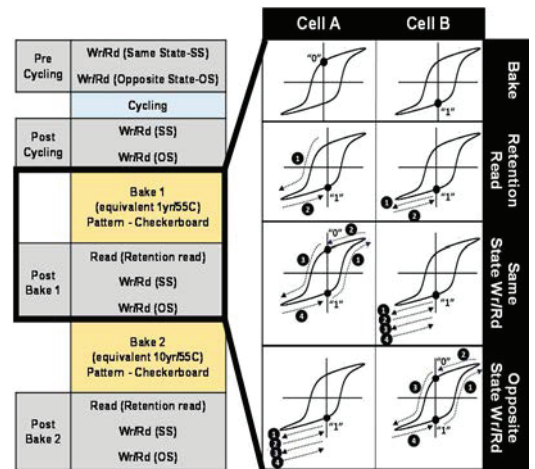


Figure 19. Block diagram of wafer level reliability (WLR) test flow with detailed schematic of bake sequence showing "same state" and "opposite state" reads.

	Read / Write Endurance	Data Retention
DRAM	$> 10^{15}$	Seconds
NVDRAM	$> 10^{15}$	> 10 years @ 55°C
NAND	$10^3 - 10^5$	> 5 years @ 55°C

Figure 22. Comparing NVDRAM endurance and retention metrics to conventional memory technologies. NVDRAM achieves DRAM-like endurance while surpassing NAND data retention.